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Vulnerability Assessment & Scoring Report

Janhavi Pawar

2124UCSF1074

Target Virtual Machine

Metasploitable2

Objective

To perform a vulnerability assessment on a vulnerable virtual machine by identifying vulnerabilities, distinguishing true and false positives, locating their CVE/CWE information, calculating CVSS scores, and performing threat modelling on a login page using the STRIDE and DREAD methodologies.

Tools Used

Nessus Essentials (or OpenVAS/Greenbone)

NVD (National Vulnerability Database)

CVE Details

CVSS Calculator

Web Browser

STRIDE & DREAD Framework

Methodology

Scanned the Metasploitable2 virtual machine using Nessus/OpenVAS.

Reviewed the scan report to identify vulnerabilities.

Distinguished true positives from false positives.

Searched the identified CVEs in the NVD database.

Recorded associated CWE IDs and CVSS scores.

Performed STRIDE and DREAD threat modeling on the DVWA login page.

Results

CVE/CWE Analysis

Vulnerability 1

Name: VSFTPD Backdoor

CVE: CVE-2011-2523

CWE: CWE-306 (Missing Authentication)

CVSS Score: 10.0 (Critical)

Description:

A backdoor present in VSFTPD version 2.3.4 allows unauthorized remote access to the server.

Vulnerability 2

Name: Samba Remote Code Execution

CVE: CVE-2007-2447

CWE: CWE-94 (Code Injection)

CVSS Score: 10.0 (Critical)

Description:

A flaw in Samba allows remote attackers to execute arbitrary commands.

STRIDE Threat Model (Login Page)

DREAD Analysis

Total Score

$(8 + 9 + 8 + 9 + 8) / 5 = 8.4$

Risk Level: High

Observations

Multiple critical vulnerabilities were identified during scanning.

The VSFTPD backdoor and Samba RCE were confirmed as true positives.

Some informational findings (such as version disclosure) may be considered false positives depending on the environment.

The login page is susceptible to several STRIDE threats, primarily Spoofing and Information Disclosure.

Conclusion

The vulnerability assessment successfully identified several critical weaknesses within the vulnerable virtual machine. CVE and CWE information was obtained from public vulnerability databases, and CVSS scores were used to assess severity. STRIDE and DREAD threat modeling highlighted key security risks associated with the login page and demonstrated the importance of implementing secure authentication, logging, and transport protection.